MIL-M-27274D (USAF)
27 January 1972
SUPERSEDING
MIL-M-27274C (USAF)
25 February 1964

MILITARY SPECIFICATION
MASK, OXYGEN MBU-5/P

1. SCOPE.

1.1 This specification covers one type of pressure breathing oxygen mask designated MBU-5/P and furnished in the following sizes (see 6.2):

Short Narrow	Long Narrow
Regular Narrow	Regular Wide

2. APPLICABLE DOCUMENTS.

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-1183	Plastic Molding Material, Acrylonitrile-butadiene-styrene (ABS), Rigid
BB-A-1034	Air, Compressed, for Breathing Purposes
BB-N-411	Nitrogen, Technical
UU-P-268	Paper, Kraft, Untreated, Wrapping
ZZ-R-765	Rubber, Silicone: Low and High Temperature and Tear Resistant
PPP-B-576	Box, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-585	Box, Wood, Wirebound
PPP-B-591	Box, Fiberboard, Wood-Cleated
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof for packaging

Military

MIL-P-116	Preservation, Methods of
MIL-C-5040	Cord, Nylon
MIL-T-8363	Tape and Webbing, Textile, Woven, Nylon
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible
MIL-O-27210	Oxygen, Aviator's Breathing, Liquid and Gas
MIL-V-27296	Valve, Oxygen Mask, Combination Inhalation and Exhalation

FSC: 1660

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STANDARDS

Federal

FED-STD-595	Colors
FED-STD-601	Rubber; Sampling and Testing

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for Selection of
MIL-STD-831	Test Reports, Preparation of
MS22064	Clamp, Oxygen Hose
MS27796	Connector-Bayonet, Three Pin, Oxygen Mask
MS27980	Fastener, Snap, Style 2
MS27983	Fastener, Snap, Style 4

DRAWINGS

Air Force

60D3516	Face Form-Oxygen Mask, Size Short Narrow
60D3519	Face Form-Oxygen Mask, Size Regular Narrow
60D3522	Face Form-Oxygen Mask, Size long Narrow
60D3525	Face Form-Oxygen Mask, Size Regular Wide
64D1676	Cover-Dust, MBU-5/P Oxygen Mask

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D624-54 Tear Resistance of Vulcanized Rubber

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

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3. REQUIREMENTS.

3.1 Qualification. Masks furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 Selection of specificaitons and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

*3.3 Materials. Materials shall conform to the applicable specifications and shall be as specified herein. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended. Materials that will come in contact with oxygen shall be entirely free from oil, grease, or any other combustible materials.

*3.3.1 Silicone-rubber material. Silicone rubber shall be used in the facepiece, the delivery tube, and the microphone cable guides of the oxygen mask and shall conform to Class 3a, Grade 50 of ZZ-R-765, except that the minimum brittle point shall be -56°C and the Shore A hardness shall be 45 $\pm$ 5. It shall be smooth and free from flash, foreign materials, or imperfections. The silicone rubber shall be post-cured in a well ventilated forced air oven. The date (by year and quarter) the silicone rubber was cured shall be molded on the facepiece and the delivery tube of the mask.

*3.3.2 Plastic materials. The hardshell shall be fabricated by injection molding from acrylonitrile-butadiene-styrene molding plastic conforming to Type I of L-P-1183. The color of the hardshell shall be the same as the color of the mask facepiece.

*3.3.3 Webbing. The hardshell webbing shall conform to Type I of MIL-T-8363. The color of the webbing shall be substantially the same as the color of the hardshell.

3.3.4 Nylon cord. The nylon cord shall conform to Type IA of MIL-C-5040.

3.3.5 Protective treatement. When materials that are subject to corrosion in oxygen, salt air, or any other atmospheric conditions likely to occur during service usage are used in the construction of the mask, they shall be protected against corrosion in the manner that in no way prevents compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

*3.4 Design and construction. The design and construction of the mask shall be in accordance with figures 1 to 9 inclusive and Drawings 60D3516, 60D3519, 60D3522, and 60D3525. Air Force drawings 60D3516, 60D3519, 60D3522, and 60D3525 should be used as general guidelines in the design of tooling.

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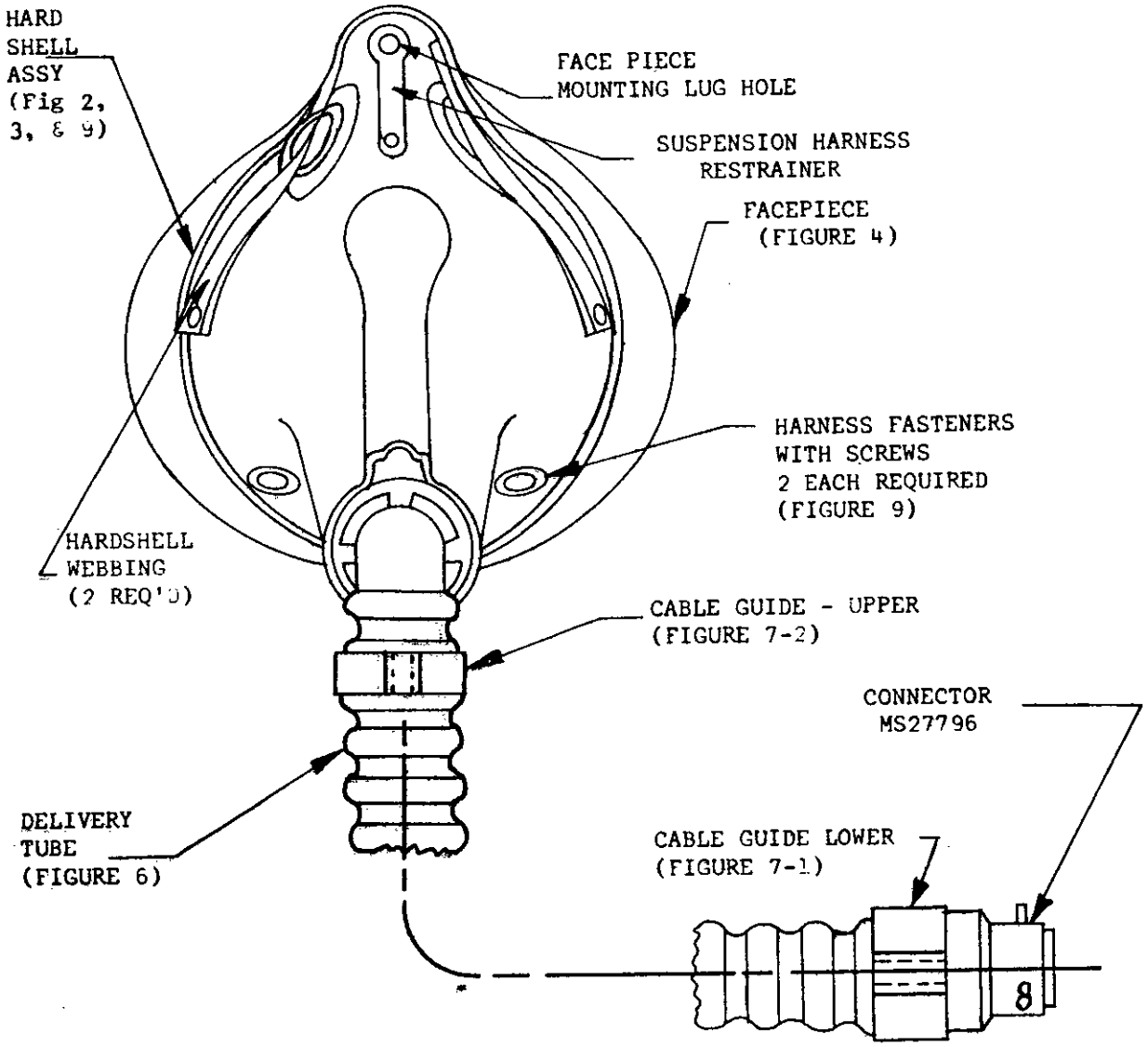
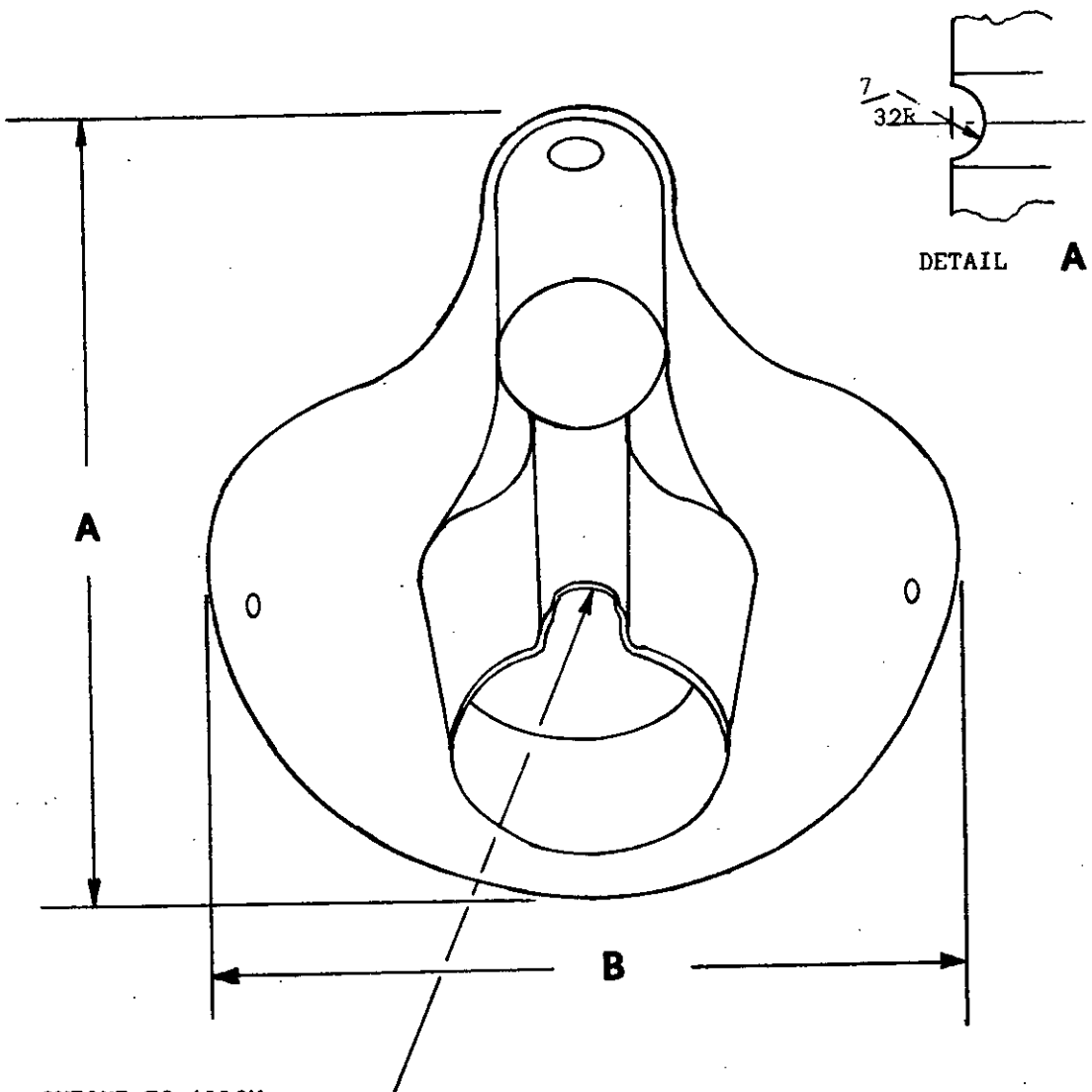


FIGURE 1 - MASK ASSEMBLY

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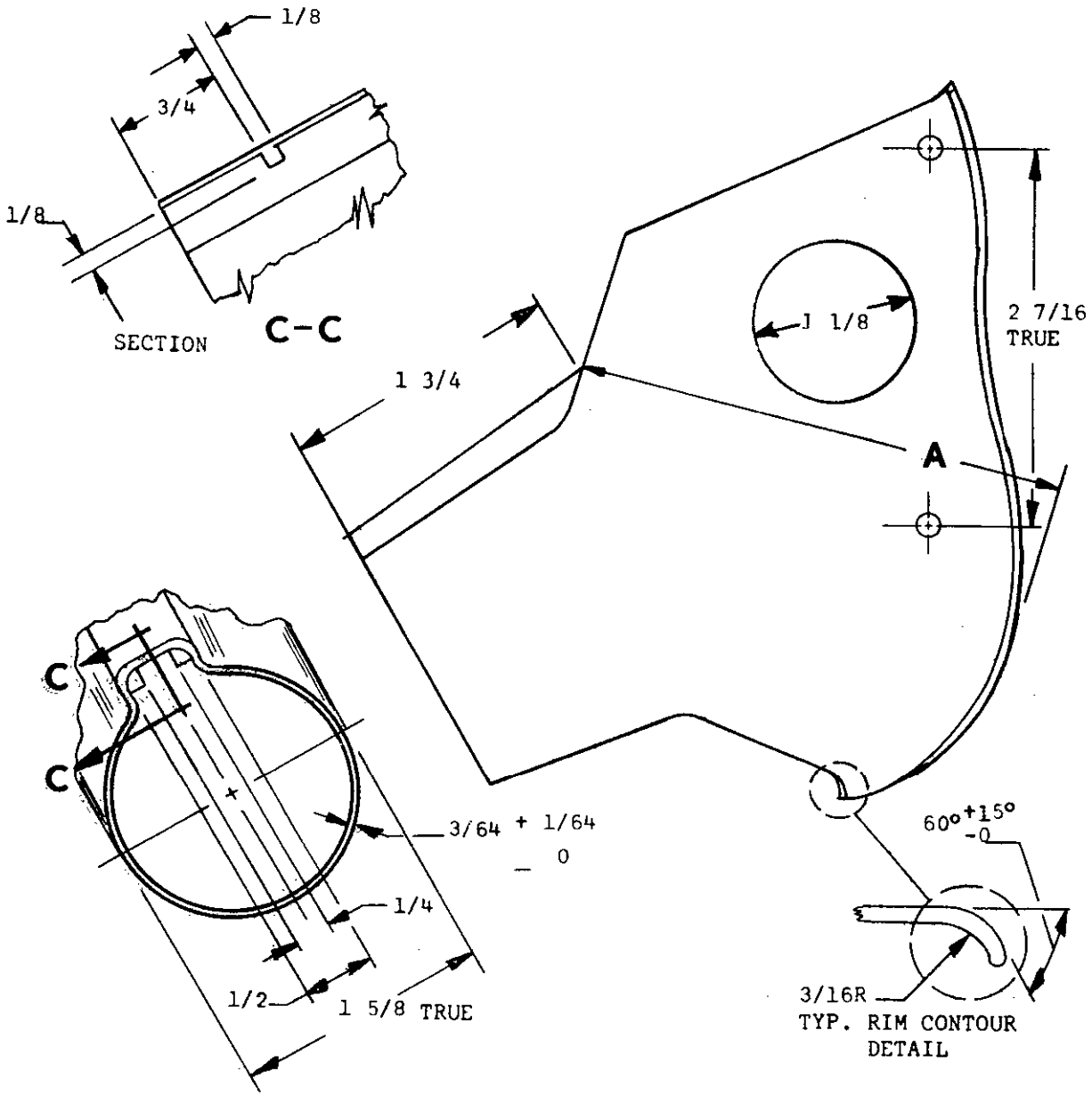
CUTOUT TO ALLOW MICROPHONE WIRE TO BE ATTACHED (SEE DETAIL "A")

DIMENSIONS IN INCHES

DIMENSION	SHORT NARROW	REGULAR NARROW	LONG NARROW	REGULAR WIDE
A	4 13/16	4 7/8	5 1/8	4 3/4
B	4 9/16	4 1/4	4 3/16	4 3/8

FIGURE 2 - MASK HARDSHELL

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DIMENSION	SHORT NARROW	REGULAR NARROW	LONG NARROW	REGULAR WIDE
A	3 1/8	3 1/8	2 15/16	2 5/8

DIMENSIONS IN INCHES

FIGURE 3 - MASK HARDSHELL DETAILS

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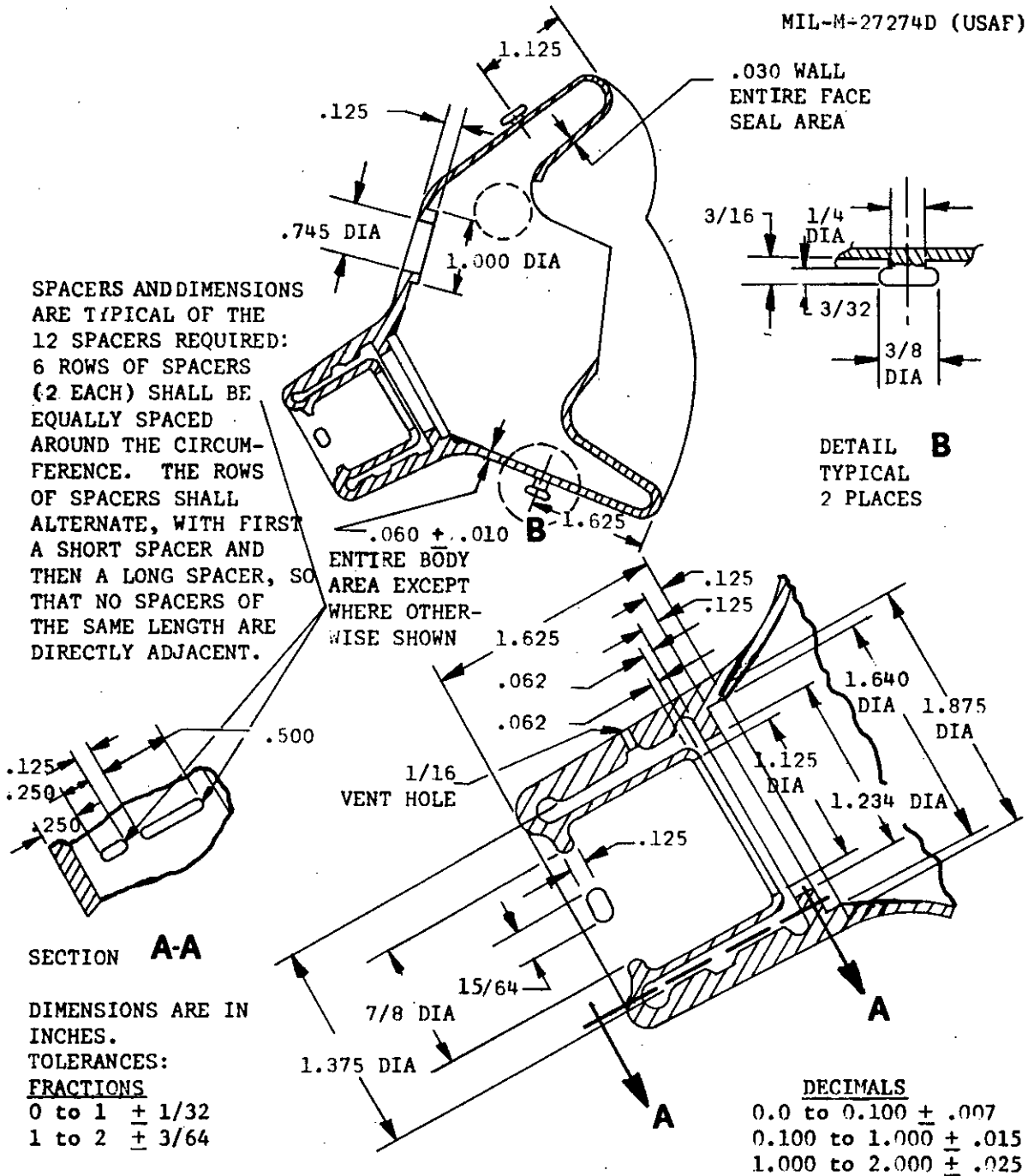


FIGURE 4 - FIXED DIMENSIONS OF THE FACEPIECE

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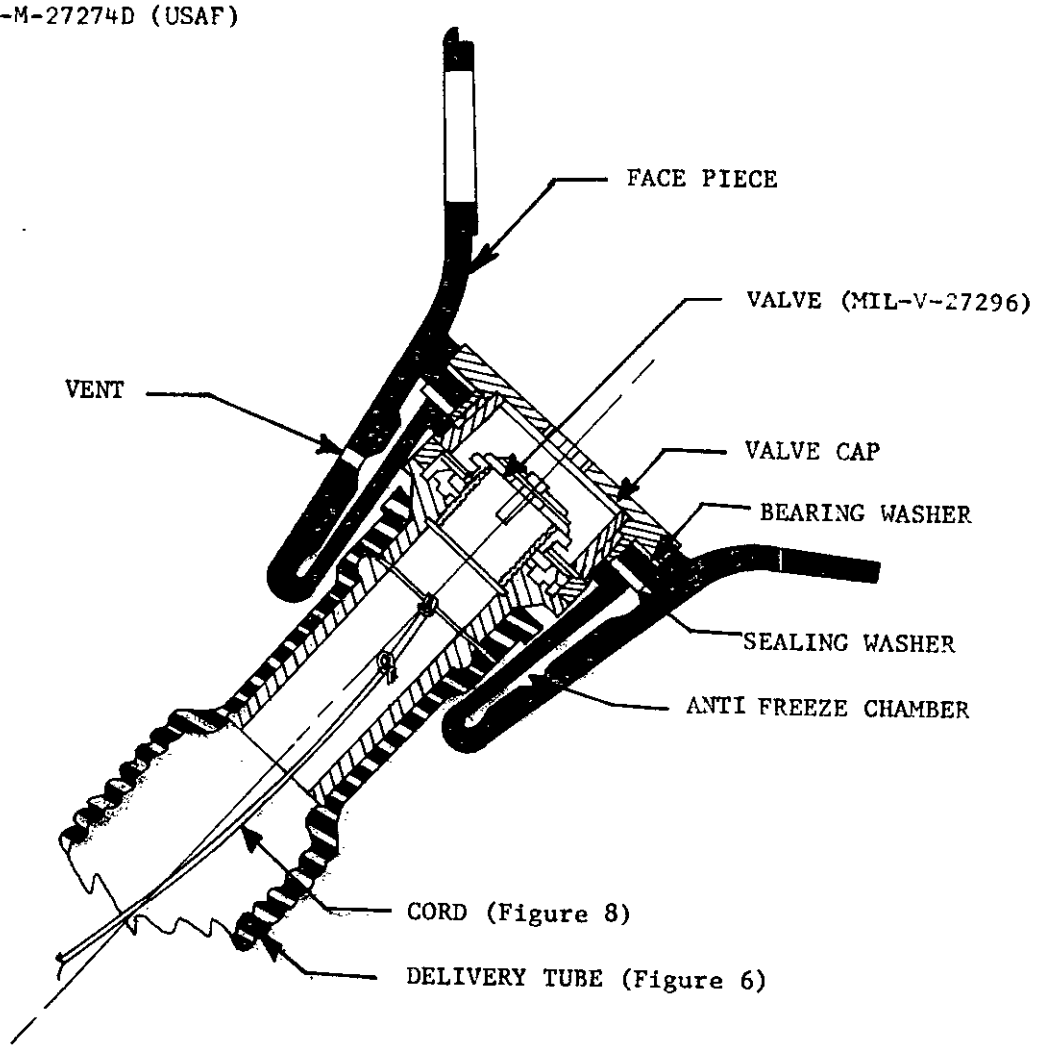
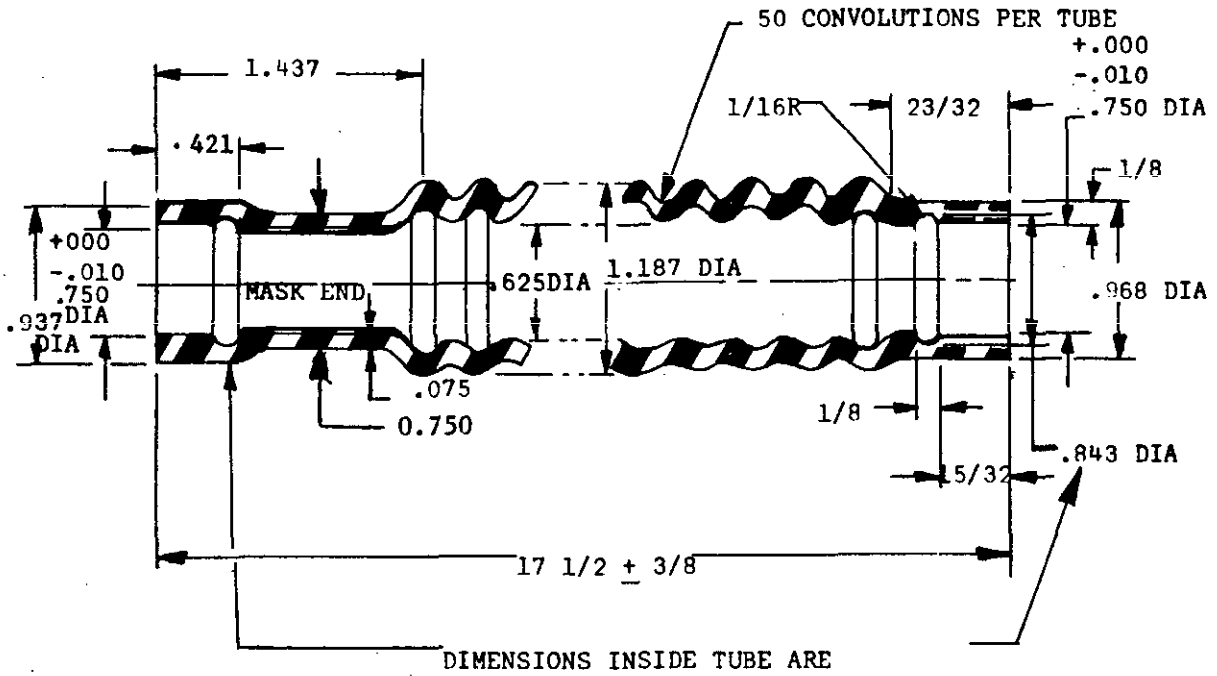


FIGURE 5 - VALVE INSTALLATION

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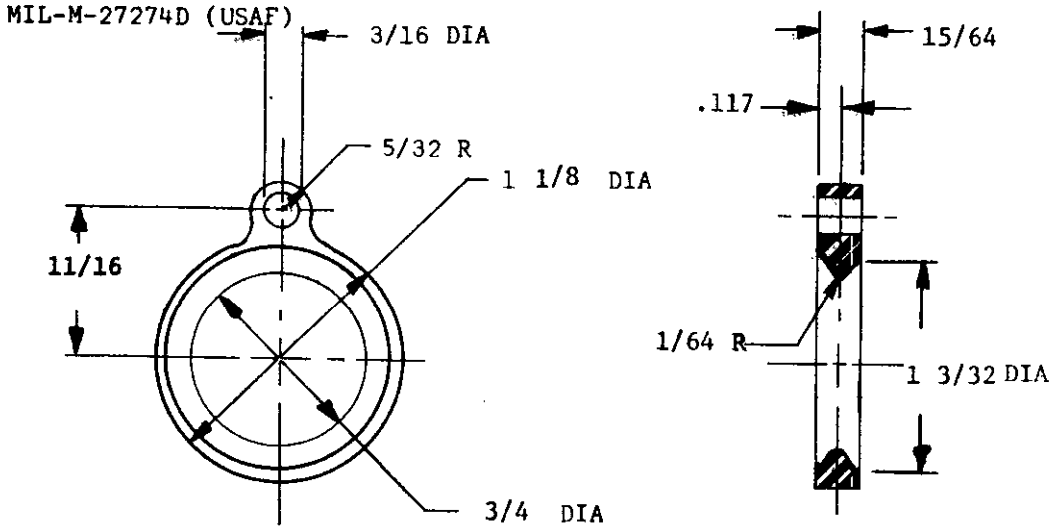


DIMENSIONS IN INCHES

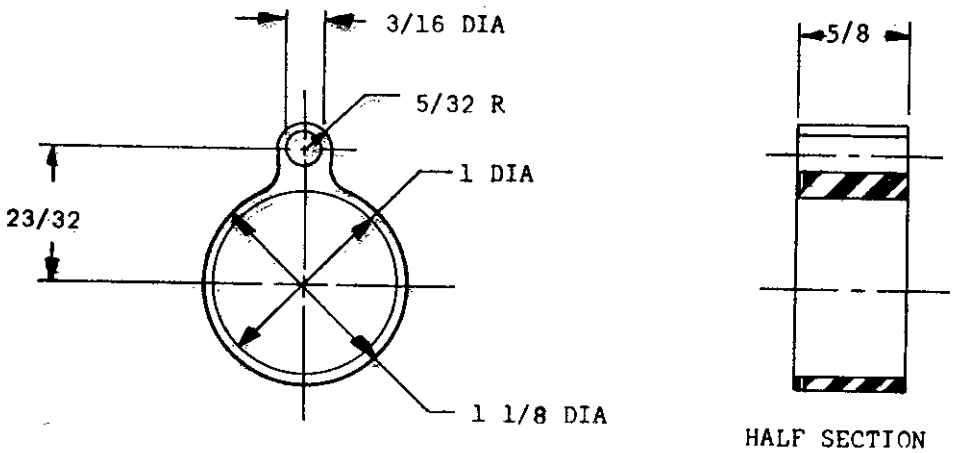
TOLERANCES ON	
FRACTIONS	DECIMALS
± 1/32	± .010

FIGURE 6 - DELIVERY TUBE

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-2 CABLE GUIDE - UPPER

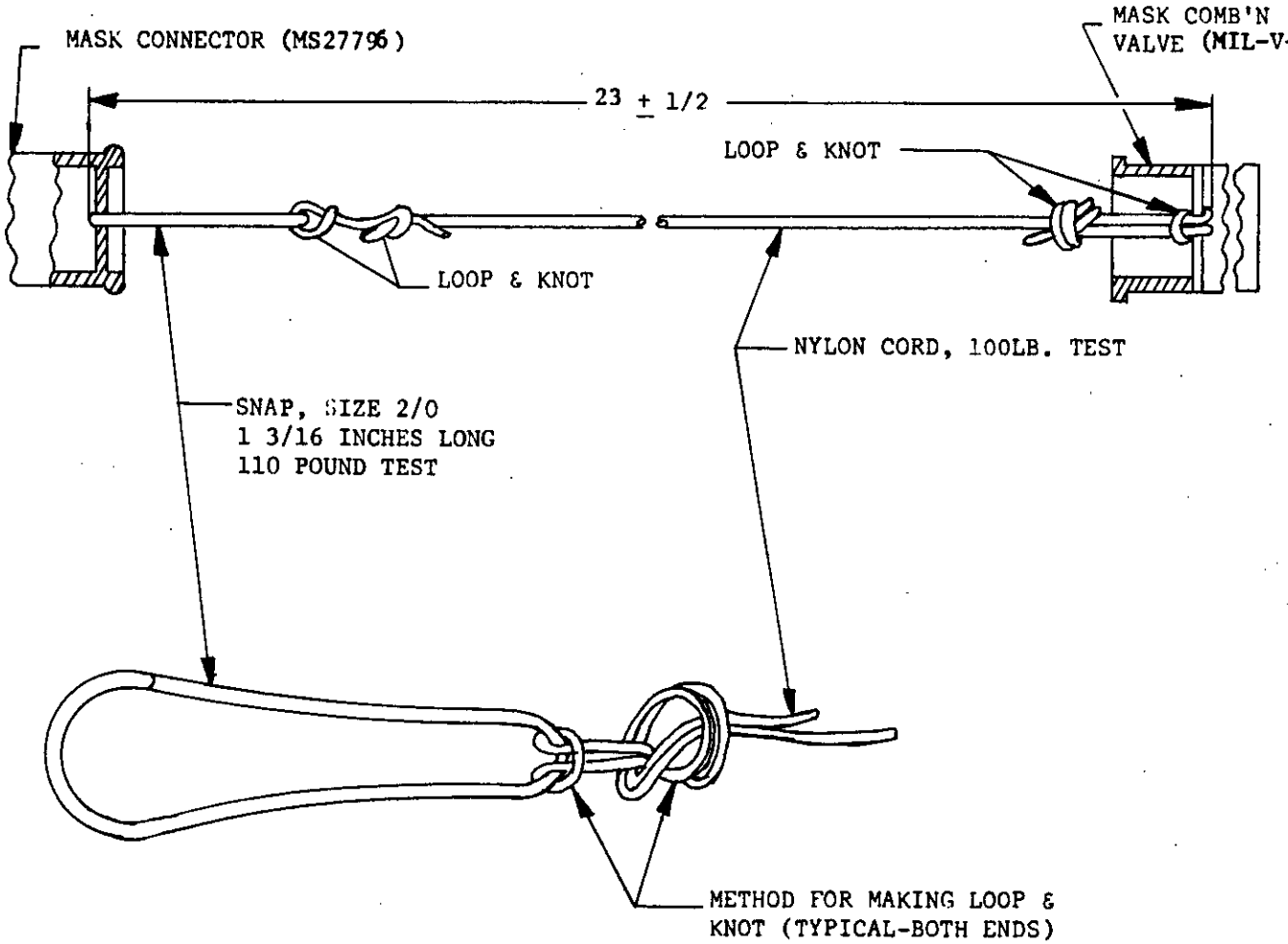


-1 - CABLE GUIDE - LOWER

DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCE FOR ALL DIMENSIONS
SHALL BE ± 1/64

FIGURE 7 - CABLE GUIDES

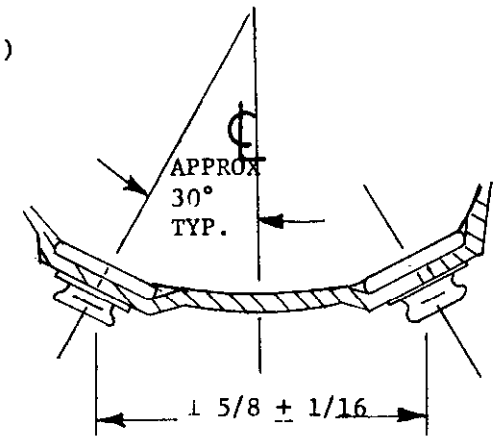
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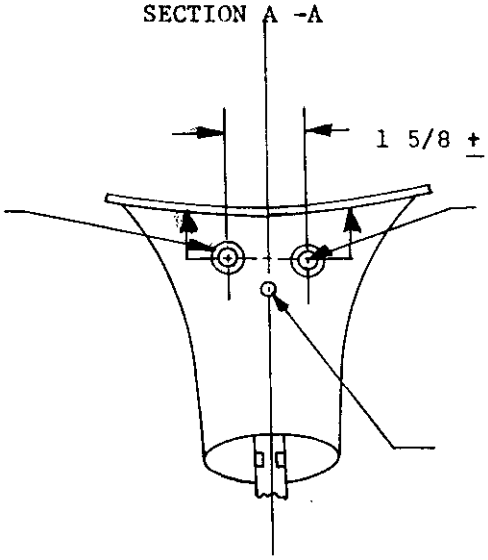
DIMENSIONS IN INCHES

FIGURE 8 - NYLON CORD & SNAP ASSEMBLY

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AS TAKEN AT OUTER SURFACE
SECTION A -A



FASTENERS
MS27980-7B
MS 27983-5
2 EACH REQ'D

OPTION:
MS 27980-7N
MS 27983-5N
2 EA. REQ'D

DIMENSIONS IN INCHES

FIGURE 9 - HARNESS FASTENER ARRANGEMENT

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3.4.1 Plug. Each mask shall have a suitable leaktight plug for the microphone flange fabricated from metal, plastic, or hard rubber.

*3.4.2 Antifreeze chamber. The antifreeze chamber and the body of the mask shall be of one-piece construction as shown in Figures 4 and 5. A 1/16-inch hole located as shown in Figures 4 and 5 shall vent the antifreeze chamber.

3.4.3 Combination valve. A combination inhalation-exhalation valve conforming to MIL-V-27296 shall be inserted in the mask as shown in Figure 5.

*3.4.4 Delivery tube. The delivery tube for the mask shall be constructed of corrugated silicone and shall conform to Figure 6. The delivery tube shall have one upper cable guide conforming to Figure 7-2 and located as shown in Figure 1.

*3.4.5 Clamp. The delivery tube shall be attached to the combination valve by means of a hose clamp conforming to Sierra Engineering Company, Sierra Madre, California, part number 450-134A or approved equal. Clamp assembly pliers conforming to Sierra Engineering Company, part number 450-801 may be used to attach the clamp.

*3.4.6 Connector. A connector conforming to MS27796 shall be attached to the delivery tube by means of a hose clamp conforming to MS22064-5 or other approved hose clamps. A lower cable guide conforming to Figure 7-1 shall cover this hose clamp and shall be located as shown in Figure 1.

*3.4.7 Limited stretch cord. A nylon cord and snap assembly conforming to Figure 8 shall be installed inside the delivery tube by attaching a nylon loop to the combination valve retention pin and a metal snap to the connector retention pin. The snap shall be Pacific Marine Supply Company, Seattle, Washington, size No. 2/0 (1-3/16 inches in length, 110-pound test), or equal, and shall be attached to one end of the nylon cord as shown in Figure 8.

*3.4.8 Hardshell assembly. The hardshell assembly shall consist of the hardshell including the installed hardshell webbing, harness fasteners with screws, and suspension harness restrainer. The hardshell assembly shall conform to Figure 1 and Figure 9.

*3.4.8.1 Hardshell. The hardshell for the mask shall closely follow the contour of the outside of the facepiece and shall be designed to assure that all of the performance requirements specified herein are met. The edges of the hardshell nearest the user's face shall be flared, and all sharp edges shall be rounded. Dimensions given in Figures 2 and 3 may be used as general guidelines in the design of the hardshell.

*3.4.8.2 Hardshell webbing. The webbing shall be riveted to the hardshell as shown in Figure 1 by 1/8-inch tubular rivets having a minimum head diameter of 9/32 inch.

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*3.4.8.3 Harness fasteners. Fasteners conforming to MS27980 and MS27983 shall be located in the hardshell as shown in Figure 9. Two machine screws (3-56NF-3A by 0.375 inch long) shall secure the harness assembly to the hardshell. These machine screws shall be provided as part of the hardshell assembly.

*3.4.8.4 Suspension harness restrainer. A suitable device as shown in Figure 1 shall be provided for restricting the lateral movement of the harness assembly when the mask is used with a continuous single strap suspension harness assembly. The device shall be provided as a component of the hardshell assembly and shall be approved by the procuring activity.

3.4.9 Dust cover. The mask shall be equipped with a dust cover to prevent dust, dirt, and foreign objects from entering the mask cavity when the mask is not in use. The cover shall be made of durable, nonbrittle, lightweight plastic material contoured to rest around the face edge of the hardshell and to be retained on the hardshell with an elastic webbing around the front edge of the hardshell at the microphone contour. The configuration of the cover shall conform to 64D1676. The cover shall not rest on the face piece contour between the seal area and the body of the facepiece.

3.5 Performance.

*3.5.1 Mask leakage. The inboard leakage shall not exceed 100 milliliters per minute when tested in accordance with 4.6.2.

*3.5.2 Combination valve installation, inlet performance (low suction). The total leakage through the combination valve inlet portion shall be 15 milliliters per minute or less when tested in accordance with 4.6.3.

*3.5.3 Combination valve installation, inlet performance (high suction). The total leakage through the combination valve inlet portion shall be 150 milliliters per minute or less when tested in accordance with 4.6.4.

*3.5.4 Combination valve installation, exhalation performance. A pressure of 20 millimeters of mercury in the facepiece and a pressure between 15 and 19.9 millimeters of mercury in the inlet tubing shall cause the exhalation portion of the valve to open when tested in accordance with 4.6.5.

*3.5.5 Combination valve installation, exhalation resistance. The exhalation resistance shall not be greater than 4.67 millimeters of mercury with a flow of 100 liters per minute when tested in accordance with 4.6.6.

*3.5.6 Delivery tube leakage. The delivery tube shall not leak when subjected to an internal air pressure of five pounds per square inch and tested in accordance with 4.6.7.

*3.5.7 Delivery tube strength. The delivery tube shall not tear or separate when tested in accordance with 4.6.8.

*3.5.8 Tear resistance. The silicone rubber shall have a minimum tear resistance of 150 pounds per inch when tested in accordance with 4.6.9 and ASTM Standard D624-54.

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*3.5.9 Hardness. The Shore-A-durometer hardness of the silicone rubber shall be 45 ± 5 when tested in accordance with 4.6.10.

*3.5.10 Toxicity. The silicone rubber shall not contain any chemicals or materials that have toxic properties in accordance with 4.6.11.

*3.5.11 Subjective use. The silicone rubber shall not produce any objectionable odors, shall not be tacky, and shall not have any other properties that cause discomfort or affect wearability. The mask shall have no detrimental or objectionable performance characteristics when evlauated in accordance with 4.6.12.

*3.5.12 Hardshell flexibility. The mask hardshell shall withstand a decrease in width (as measured by dimension B of Figure 2) of 1 1/2 inches without visible cracking or fracture when subjected to compressive loading in accordance with 4.6.13.

*3.6 Color. The color of the hardshell, webbing, and all silicone rubber used in the oxygen mask assembly shall be green, approximately matching color numbers 14158, 24158, and 34158 of FED-STD-595.

3.7 Part numbering if interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

*3.8 Identification of product. The mask assembly shall be marked with the following specific information in accordance with MIL-STD-130. The Federal Stock Number shall be omitted from the nameplate.

Mask, Oxygen MBU-5/P
Mask Size*
MIL-M-27274D (USAF)
Manufacturer*
Date of Manufacture (Quarter and Year)*

*The manufacturer shall include the applicable information.

3.8.1 Tubing. The mask end of the tubing shall be identified by molding MASK END into the rubber. The date of manufacture (Quarter and Year) shall be molded on either end of the tubing.

*3.9 Workmanship. The mask shall be constructed in accordance with good commercial practice, and shall be free from any defects which might affect appearance or functionability. The mask shall be clean and dry, free from oil, grease, and other foreign matter. All metal and plastic parts shall be smooth and free of sharp edges. All excess flash shall be properly trimmed. The face piece and its related parts shall be smooth and free from blisters, porous areas, cracks, abrasions, bubbles, and foreign material.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Test conditions.

4.3.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77° ± 18° F, and at a relative humidity of 80 per cent or less. Where tests are made with atmospheric pressure or temperature substantially different from these values, proper allowance shall be made for the change in instrument reading.

*4.3.2 Gas. The gas used in qualification testing shall be oxygen conforming to Type I of MIL-O-27210. The gas used in all other testing shall be oxygen conforming to Type I of MIL-O-27210, water pumped nitrogen conforming to Type I, class 1, grade B of BB-N-411, or compressed air conforming to grade A or C of BB-A-1034. If nitrogen or air is used, appropriate density correction factors shall be applied to the flowmeter used.

4.4 Qualification inspection.

4.4.1 Test sample. The test sample shall consist of one mask of each size, three extra hardshells of each size, and test sheets of silicone material. The test sheets shall consist of four sheets that are 6 inches by 6 inches by 1/16 inch. An additional sample of each size of facepiece and four test sheets shall be required for each change of silicone rubber compound formulation used in the fabrication of the facepiece. An additional sample of three hardshells of each size shall be required for each change of plastic material used in the fabrication of the hardshell. The test sample shall be representative of the production equipment and shall be identified with the manufacturer's part number, the compound number of the manufacturer of the silicone material, and such other information as required by the qualifying activity.

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4.4.2 Reports and test sample for the qualifying activity. The following reports and samples shall be furnished to the qualifying activity:

a. Three copies of a test report prepared in accordance with MIL-STD-831. The test report shall include the results of all tests and a detailed statement of compliance or noncompliance with each requirement of this specification, identified by the applicable paragraph number.

b. Three copies of a silicone-rubber report containing the following information:

(1) Compound number and name of manufacturer

(2) Composition of compound, including fillers and catalysts, giving parts by weight of each constituent.

(3) Mixing and curing instructions.

(4) Any additional information the manufacturer desires to furnish for better identification of the material or to show that the material has desirable characteristics for oxygen mask usage.

c. The samples that were tested.

4.4.3 Qualification tests. The qualification tests shall consist of all tests described under 4.6.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual tests.

b. Sampling plans and tests.

4.5.1 Individual tests. Each oxygen mask shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Mask leakage

c. Combination valve installation, inlet performance (low suction)

d. Combination valve installation, exhalation performance

e. Delivery tube leakage.

4.5.2 Sampling plans and tests.

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4.5.2.1 Sampling plan A. Two masks of each size specified in the contract or order shall be selected at random from each lot of each size of 1,000 masks (or fraction thereof) and shall be subjected to the following tests as described under 4.6:

a. Combination valve installation, inlet performance (high suction).

b. Combination valve installation, exhalation resistance.

c. Delivery tube strength.

4.5.2.2 Sampling plan B. One mask facepiece of each size specified in the contract or order shall be selected at random from each batch of silicone rubber compound received from the manufacturer. Test sheets fabricated in accordance with 4.4.1 shall also be made from each batch. The test sample shall be subjected to the following tests as described under 4.6:

a. Tear resistance.

b. Hardness

4.5.2.3 Sampling plan C. If hardshells are molded in an essentially continuous operation, three hardshells shall be selected at random from each day's production of each mold cavity and subjected to the test described in 4.6.13. If hardshells are not molded in an essentially continuous operation, three mask hardshells of each size of masks specified in the contract or order (fabricated from each batch of material received from the manufacturer and molded under the same conditions and at substantially the same time) shall be selected at random and subjected to the test described in 4.6.13.

4.5.3 Reject criteria. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined and corrected. Individual tests may be continued pending the investigation of a sampling test failure. However, final acceptance of items on hand or later produced shall not be made until it is determined that the items meet all the requirements of this specification. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods.

4.6.1 Examination of product. Each oxygen mask shall be carefully examined to determine conformance to the requirements of this specification with respect to materials, construction, workmanship, dimensions, and marking.

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*4.6.2 Mask leakage. Each mask (with combination valve installed correctly and the microphone plug in place) shall be placed on a suitable test stand and adjusted so there is no leakage around the face seal. The inboard leakage shall not exceed 100 milliliters per minute when a suction of 13 millimeters of mercury is applied within the facepiece of the mask with the free end of the mask tubing plugged.

*4.6.3 Combination valve installation, inlet performance (low suction). The inlet valve portion shall be completely dry throughout this test. The inlet valve shall be inspected for imperfect seating of the flapper on the seat and leaks around the valve seat by applying and maintaining a suction of 0.19 millimeters of mercury at the free end of the delivery tube. The total leakage through the combination valve inlet portion shall not be more than 15 milliliters per minute. If necessary, a suitable means shall be used to prevent collapse of the hose.

*4.6.4 Combination valve installation, inlet performance (high suction). This test shall conform to 4.6.3 except that a suction of 22.4 millimeters of mercury shall be used. The leakage through the combination valve inlet portion shall not exceed 150 milliliters at this suction.

*4.6.5 Combination valve installation, exhalation performance. Each mask shall be placed on a suitable test stand and all suction and pressure from the previous test exhausted. A pressure of 20 millimeters of mercury shall be applied at the mask tubing. The pressure at the mask tubing shall then be gradually reduced, causing the inlet valve to close and causing a pressure differential to be applied across the exhaltion valve. The mask exhalation valve shall open with a pressure in the tube between 15 and 19.9 millimeters of mercury. If an extreme pressure is encountered, the inlet valve shall be rechecked for leakage. If the valve is properly seated and the mask still fails to function properly, the valve shall be replaced and the test repeated.

*4.6.6 Combination valve installation, exhalation resistance. The mask (with the combination valve installed) shall be placed on a suitable test stand. At an exhalation flow of 100 liters per minute, the pressure inside the mask shall not exceed ambient pressure by more than 4.67 millimeters of mercury.

*4.6.7 Delivery tube leakage. One end of the delivery tube shall be plugged and an air pressure of 5 pounds per square inch shall be applied to the other end. The tube shall be submerged just under the surface of water. Bubbles rising from the hose after 10 to 15 seconds of submersion shall be cause for rejection.

4.6.8 Delivery tube strength. One end of the tubing shall be attached to the combination valve mounted on the mask body. The other end shall be connected to a fitting simulating a connector conforming to MS27796. Clamps for the connections shall be of the type specified herein. One of the end connections shall be attached to a support. A weight of 63 pounds $\pm$ 1 pound shall be attached to the other connection so that a straight pull is exerted on the tubing for a period of 3 seconds $\pm$ 2 seconds. The tubing shall not

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tear or separate from the end connections. The nylon antistretch cord shall be attached during this test. Upon completion of this test the delivery tube shall be subjected to the test of 4.6.7.

4.6.9 Tear resistance. The tear resistance of the test sheets and sections of rubber cut from the sealing area of the facepiece shall be determined in accordance with ASTM D624-54, die B. Minimum tear resistance shall be 150 pounds per inch.

*4.6.10 Hardness. The hardness of the test sheets and sections of rubber cut from the sealing area of the facepiece in both the nose and check areas shall be determined in accordance with method 3021 of FED-STD-601, and shall have a shore A hardness of 45 ± 5.

4.6.11 Toxicity. Toxicity shall be determined from the chemicals used in formation of the silicone rubber compound.

*4.6.12 Subjective use. Objectionable odors, tackiness, and objectionable or detrimental performance characteristics shall be determined by subjective evaluation. The mask shall be used by human subjects at altitudes from ground level to 50,000 feet, at temperatures from -40°F to +160°F or any combination of such altitude and temperature for periods up to 2 hours. Subjective use tests are optional tests to be conducted at the discretion of and by the procuring activity when the mask appears to have performance characteristics or other properties which might be detrimental or objectionable for human use.

*4.6.13 Hardshell flexibility. The hardshell shall be placed between the heads of a suitable compression testing machine with the heads contacting the sides of the hardshell at the point of maximum width. The hardshell shall be oriented so that dimension B of figure 2 is parallel to the direction of movement of the heads. A suitable means of measuring the decrease in hardshell width under load shall be provided. The hardshell shall be deformed until failure or until the required decrease in width is attained, whichever occurs first. The rate of head travel shall be from 0.20 to 0.25 inch per minute. The hardshell shall withstand a decrease in width of at least 1 1/2 inches without visible cracking or breaking.

*4.6.13.1 Special tool. A special compression tool for applying the load to the hardshell shall be used to minimize any effects caused by shifting of the test machine heads. This tool shall be constructed so that loading is truly parallel to dimension B of figure 2 and is applied through tool surfaces that are flat, parallel, and perpendicular to dimension B of figure 2.

*5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.

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5.1.1 Level A. The mask shall be placed in the container in such a manner as to prevent distortion, bending, and flattening of any part of the mask. The mask shall be packaged in accordance with method IA-15 of MIL-P-116. An overwrap of 30-pound paper conforming to UU-P-268, or equal, shall be provided to protect the barrier materials from abrasion. The overwrap shall be secured with tape or any other suitable means.

5.1.2 Level C. The mask shall be packaged in a manner to prevent distortion of the mask and to afford adequate protection against deterioration and physical damage during shipment from the supply source to the receiving activity.

5.2 Packing.

5.2.1 Level A. The mask shall be packed in an overseas type shipping container conforming to PPP-B-576, PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, or PPP-B-636. As far as practical, the container shall be of uniform shape and size, shall be of minimum cube and tare consistent with the protection required, and shall contain identical quantities. The containers shall be provided with a case liner conforming to MIL-L-10547 and shall be sealed in accordance with the appendix thereto. The case liner shall not be required when the container conforms to type I or II, class 2 or 3 of PPP-B-636, and is sealed at all joints and seams with tape conforming to PPP-T-60. The container shall be limited to a gross weight of approximately 200 pounds and shall be sealed and strapped in accordance with the applicable specification and appendix thereto.

5.2.2 Level B. The mask shall be packed in a domestic type shipping container conforming to PPP-B-576, PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, or PPP-B-636. As far as practical, the container shall be of uniform shape and size, shall be of minimum cube and tare consistent with the protection required, and shall contain identical quantities. The gross weight of each container shall be limited to approximately 200 pounds and shall be closed in accordance with the applicable specification and appendix thereto.

5.2.3 Level C. The mask shall be packed in a manner to insure acceptance by the carrier and safe delivery to destination. Containers shall conform to Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The oxygen mask covered by this specification is for use in dispensing gaseous oxygen from pressure demand regulators to aircrew members engaged in high altitude flying.

6.2 Ordering data. Procurement documents should specify the following:

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a. Title, number, and date of this specification
b. Size or sizes required (see 1.1)
c. Qualification requirements
d. Levels of preservation, packaging, and packing required (see section 5).

*6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City Air Materiel Area (MMEO), Tinker AFB, Okla 73145 and information pertaining to qualification of products may be obtained from that activity.

NOTE: The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 71

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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